Unsupervised Value Discovery: Locating the Value Structures of an Agent from Passive Observation by their Empowerment–Plasticity Asymmetry

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Abstract

Unsupervised agent discovery (UAD) locates agents in raw multivariate time series by finding variable sets that approximately satisfy a Markov-blanket condition against their surroundings. We ask a question one level inward: within a discovered agent, which parts are its *value structures*, the components of its informational structure that correspond to its goals, objectives, or desires? We propose that this role has a directional information-theoretic signature: values are the components that most strongly drive the world while being least driven by it, maximally empowered and minimally plastic. This inverts a familiar AI safety assumption that a system first develops values and then acts to make them empowered and stable; we hypothesize instead that whatever is empowered and stable thereby *serves as* the system's values, and that this structural property is passively detectable. We operationalize the hypothesis, as a *value signature* on directed-information scores gated by procedure-mirroring noise floors, and evaluate it in three experiments, using worlds built on an existing UAD benchmark. These show the following. (i) In a single-agent world, the value signature identifies a planted goal, and only it, when using measurement conventions that can see through the world's XOR-style composition. (ii) Across four more complex worlds, including an eight-agent, 49-variable colony world, only a block-level measurement convention, which fuses discovered agents and offers every block as a conditioning 'key', recovers the planted goals. (iii) Against adversarial impostors, a fast environment-driven 'captured goal' is refused but two others proved more difficult. One, a slow history-triggered captured goal, was wrongly identified at first, although follow-up experiments showed this was due to our choice of parameters rather than to any limit of passive observation. The other, a noiseless goal twin, was passively indistinguishable in principle and could only be identified with an interventional yardstick generalizing the benchmark's goal-flip operation, showing the limits of passive detection. We report limitations (single-variable values; lag-1 conditioning) and directions for further work.

1 Introduction

The standard route to identifying an agent's objectives runs through its behaviour: observe choices, infer the utility or reward that rationalizes them [4, 6]. This route has well-known difficulties,

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non-identifiability, dependence on rationality assumptions, and, one that is rarely foregrounded, an agent may *conceal* its objectives, for instance by injecting noise into its decisions, so that the behavioural signal is deliberately weakened. This paper explores a complementary route. Rather than inferring values from what a system does, we look for its value structures in the *informational structure* of the system itself: which internal component most stably drives behaviour and impact? If such a component can be located structurally, it can be isolated and studied directly, recovering the signal that behavioural inference alone would lose.

Our starting point is unsupervised agent discovery (UAD) [8], which finds agents in raw time series as variable clusters that approximately satisfy a Markov-blanket condition against their environment [2, 3]. UAD locates agents' *boundaries*. We ask what *drives* them, and specifically what plays the role of a discovered agent's values. We deliberately say value *structure* rather than value *function*: locating the component that plays the value role does not tell us what that component does or prefers, and we have not built tools for that.

The hypothesis: inverting the classic AI drives argument. Omohundro's *Basic AI Drives* [5] argues that an advanced system's utility function "will be precious" to it, so the system will act to make that function maximally influential and maximally protected from change. In the vocabulary of Abel et al. [1], it would seek to make them maximally *empowered* and minimally *plastic*. Omohundro's direction of explanation runs from values to structure. We hypothesize the reverse: whatever component of a system is maximally empowered and minimally plastic will, by that fact, *serve as* the system's values. This is why we speak of *value* discovery rather than *goal* discovery: we are identifying the parts that play the value role, whatever they are. In the simple systems studied here that part is a single planted goal bit; however, the same structural signature could, in principle, locate richer value structures in more complex systems.

The empowerment–plasticity trade-off [1] makes the hypothesis non-trivial: a system cannot be maximally both across its boundary, so adaptive systems have reason to specialize internally: developing belief-like parts skewed towards plasticity, value-like parts towards empowerment, and decision-making parts sitting between. Our claim is that this specialization is passively detectable.

Contributions. This paper contributes four things towards testing this hypothesis.

(1) An operationalization of our hypothesis, the *value signature*, with procedure-mirroring noise floors and controls, designed so a non-technical collaborator could audit every step in plain English.

(2) An instrument for unsupervised value discovery: directed-information scores computed under a menu of measurement conventions and gated by those floors. Our experiments across four worlds found a block-level convention (fuse discovered agents; offer every block as a conditioning key) that identifies every planted value at a scale where all variable-level conventions fail.

(3) A characterization of the instrument's limits by adversarial impostors, together with an interventional yardstick that generalizes the UAD benchmark's goal-flip operation and resolves a case (involving a zero-noise twin) in which passive detection failed.

(4) Several mechanism-level findings (an XOR 'cipher' that hides inputs from under-conditioned measures; the self-penalizing property of procedure-mirroring floors; and a 'twin channel' by which a disconnected value persists in the value signature through its own record).

2 The value signature

This section defines the two directed measures the instrument computes, the noise floors that every score must beat, and the value signature itself.

2.1 Directed measures

The idea is to measure, for each candidate component, how strongly it pushes on everything else and how strongly everything else pushes back on it. Concretely, let X be a candidate variable in a multivariate discrete time series and R 'the rest' under some convention. Using the smoothed plug-in conditional mutual information of the UAD implementation (Laplace $\alpha = 0.1$, nats), we define lag-1 directed flows of transfer-entropy type [7]:

$$\text{intake (push-in):} \quad \mathrm{I}_{\mathrm{in}}(X) = \mathrm{I}(X_{t+1};\, R_t \,|\, X_t)\,, \tag{1}$$

$$\text{drive (push-out):} \quad \mathrm{I}_{\mathrm{out}}(X) = \mathrm{I}(R_{t+1};\, X_t \,|\, R_t)\,. \tag{2}$$

The *polarity* $(\mathrm{I}_{\mathrm{out}} - \mathrm{I}_{\mathrm{in}})/(\mathrm{I}_{\mathrm{out}} + \mathrm{I}_{\mathrm{in}}) \in [-1, 1]$ is the fraction of a variable's traffic that is outbound; a raw difference is always reported alongside. Different conventions instantiate R differently (Section 3.1); an environment-flavoured drive toward a designated environment variable is also computed, never conditioned through the agent's action, because the value's path to the world runs through the action and conditioning on a mediator screens off precisely the flow of interest (verified analytically in Section 4).

2.2 Floors

Every score was compared to its own null distribution: the candidate's column was circularly shifted against the fixed remainder (≥ 200 shifts, pooled across seeds where registered), and the *identical* scoring procedure was re-run, including any maximization over conditioning sets, so that search-induced bias was priced into the floor. A score counted as *above floor* if it exceeded the 95th null percentile *and* cleared the null mean by $z \geq 3$ null standard deviations (the margin gate was added after a near-frozen variable was found to pass hair-thin residuals; see Section 5). We write $\mathrm{net}(\cdot) = \max(0,\ \mathrm{score} - \mathrm{floor}_{95})$.

2.3 The value signature

A variable X exhibits the value signature if (i) its drive is above floor, (ii) its total flow $\mathrm{I}_{\mathrm{in}} + \mathrm{I}_{\mathrm{out}}$ is above floor (it is *rankable*), and (iii) $\mathrm{net}(\mathrm{I}_{\mathrm{out}}) > 0$ and exceeds a minimum ratio of $\mathrm{net}(\mathrm{I}_{\mathrm{in}})$ (our experiments initially used $9\,\mathrm{net}(\mathrm{I}_{\mathrm{in}})$ but further work indicates a higher value performs better, see Section 6, although the best threshold is yet to be determined experimentally).

Clause (iii) is deliberately leak-tolerant: real components are all affected by their environment, so a value must not be disqualified for measurable intake, only for failing to be overwhelmingly a driver (a $9\times$ ratio corresponds to polarity ≥ 0.8 and sits well above the ≈ 3–$4\times$ observed for genuine mediators). 'Intake at floor' is the special case $\mathrm{net}(\mathrm{I}_{\mathrm{in}}) = 0$.

3 Experimental design

This section describes the worlds the experiments ran in, the conditions and controls attached to every run, and the menu of measurement conventions under test.

Table 1: Worlds. All colony worlds have eight agents with a neighbour's action nudging each agent's patch on $\approx 10\%$ of steps.

World	Variables	Planted structure / probe
Anchor (Exp. 1)	9	one agent with one goal; two perfect lookalike logs
Slow-meter (Exp. 2)	14	goal flipping slowed to 0.005; four causally inert meters
Deep-synergy (Exp. 2)	17	two agents, two beliefs each; parity vs. majority rule
Colony (Exp. 2)	49	eight goals; ring coupling; partitioned variant for attribution
Puppet colonies (Exp. 3)	49	one goal replaced by a *captured* goal (fast / slow-triggered)
Alias colony (Exp. 3)	51	goal twins for two agents at noise $\{0, 0.5, 1, 2, 5\}\%$

Worlds. All worlds derive from the `SyntheticHandleWorld` of the UAD handle benchmark [8]: nine binary variables per agent forming the loop belief $B \to$ action $A \to$ environment $E \to$ sensor $S \to B$, with a goal G entering the policy ($A = B \oplus G$ up to noise) and flipping by its own coin at rate 0.015; plus a distractor D shadowing E, pure noise W, and in the original world two perfect 'lookalike' logs of the sensor and action lines. The benchmark for these worlds was built by Gunnar Zarncke before this project existed and for a different purpose, so were not tuned to flatter our measure. Table 1 lists the worlds, each of which came with an audit (a change-frequency chart, a mechanism table comparing observed disagreement rates against those predicted from the designed noise, with agreement to $\lesssim 0.1\%$, and a plain-language 'story'), reviewed before any scoring.

Conditions and controls. Each world ran under three conditions: *main*; *no-core* (goal variables present but disconnected from the policy, removing the value signature); and *scramble* (every column independently time-shuffled, the count of above-floor scores must sit in the 5% chance band doubling as a calibration check on the floors). A contrast world from Zarncke's older multi-agent simulator supplied a variable named `goal_progress`, a causally inert progress meter, which the instrument needed to refuse.

Pre-registration and discipline. Success criteria (thresholds $\geq 80\%$ of 20 seeds; $\geq 90\%$ for goal-above-belief; uniqueness = no non-goal variable holding the value signature in more than 2 of 20 seeds) were locked before each sweep; every calibration amendment made before a sweep was dated in a decisions log. Two working rules shaped the analysis: when two requested measures disagreed, the disagreement was investigated and mechanistically explained rather than averaged; and conventions were run in parallel as co-equal tests rather than a primary being chosen early. Ground-truth labels were used only to evaluate outcomes, never inside any measurement.

3.1 Measurement conventions

Table 2 lists the conventions evaluated. The variable-level conventions differ in how R is instantiated. The block-level conventions first run the UAD detector to partition variables into agents (its cluster-count dial chosen by how many variables land in blocks that still pass the Markov-blanket check, as in Zarncke's own selection procedure), fuse each agent's variables into a compressed macro-state block (the 64 most frequent joint states, with the probability mass lost to compression reported), and never fuse the environment (because agent blocks respect a discovered coherence, while the environment has none to respect). Orphaned variables (those the detector drops: characteristically the goals, whose rare flips carry almost no symmetric mutual information) remain individually scoreable.

Table 2: Measurement conventions. 'Own-block' conditions on the candidate's own agent block; 'any-block' offers every block (and singleton) as a conditioning key.

Convention	Level	Rest / conditioning
pairwise	variable	sum of one-at-a-time flows
fused	variable	joint state of all other variables, both directions
fused + best-key	variable	fused intake; drive = best over no key and each single key
fused + grown keys	variable	fused intake; drive = greedy key growth (cap 2)
fused-agents own-block	block	own agent block standing in the conditioning
fused-agents any-block	block	target block's past + best single key element

For the variable-level conventions we ran four alternatives. 'Pairwise' sums one-at-a-time flows between the candidate and each other variable. 'Fused' takes the joint state of every other variable as R, in both directions. 'Fused + best-key' keeps the fused intake but, for drive, takes the best reading over no key and each single additional conditioning variable (the 'key' that decrypts XOR-style composition). 'Fused + grown keys' does the same with up to two keys, added greedily.

4 Experiment 1: does the value signature identify a planted value?

The anchor world was scored under pairwise, fused, and fused+best-key, in the world as-is and with the lookalike columns removed, on 20 seeds. Before any scoring, the estimator was verified against analytic targets (a delayed copy reads $\ln 2$ nats; a chain $X \to Y \to Z$ shows $\mathrm{I}(Z_{t+1}; X_t \mid Y_t) \approx 0$ while the unconditioned lag-2 dependence is positive, the mediation exhibit; the Marko–Massey conservation identity holds on analytic cases).

The belief-goal-action cipher. A first pass showed the action's intake reading ≈ 0.005 nats pairwise while a pre-planned fused cross-check read 0.31. Investigation, prompted by the disagreement rule, revealed the mechanism: XOR composition ($A = B \oplus G$) is a perfect cipher: each input is invisible to a measure conditioning on too little unless the other input (the 'key') is conditioned on. Measured readings matched hand-derived values to three decimals; re-timing the world did not help (tested), and a single extra conditioning variable restored the goal's drive on the environment from 0.000 to 0.441 nats. This motivated the key-based conventions and, later, the block-level ones.

Results. Table 3 summarizes the results of this experiment. Under fused and fused+best-key the goal exhibits the value signature in 100% of seeds, uniquely in $\geq 95\%$, and had a higher polarity than the belief in 100%; all controls pass; the only intruder anywhere was pure noise W in one seed in twenty, the false-positive rate the floors are designed to admit. Under pairwise the action wires share the value signature in 20/20 seeds (as pre-registered) and, with honest floors, the goal's own pairwise signal is indistinguishable from nothing: pairwise measurement cannot tell a value from a wire in a world with encrypted composition, and was retired, with its failure mechanism documented. Removing the lookalike columns changed no verdict, so raw data needs no pre-cleaning. The UAD detector run on the same trace recovered the agent's body as one blanket-valid cluster (leakage 0.001 against tolerance 1.0) but drops the goal as an isolated singleton: boundary-finding and value-finding are complementary axes.

Table 3: Experiment 1 (anchor, 20 seeds). Rates are fractions of seeds. Uniqueness = nothing but G holds the value signature; shown as-is / lookalike-free where the two configurations differ. GP = `goal_progress` refused; n/a = not run.

Convention	G signature	unique	$G > B$	controls	GP refused
fused	1.00	0.95 / 1.00	1.00	pass	n/a
fused + best-key	1.00	0.95	1.00	pass	0.70
pairwise	0.05	0.00	0.05	pass	1.00

The `goal_progress` residual and the floor standard. Best-key refused `goal_progress` in only 70% of seeds, where the registered expectation was refusal outright (the contrast existed solely to be refused). At first we thought this was because of slowness: cumulative meters could have autocorrelation-inflated shuffle floors, letting the witness-like drive clear the floor while the intake read as zero. However, the stored records indicated that the real reason was a *near-frozen* column (change rate ≈ 0.000) whose every score sat at its own unstable floor, plus initial transients from an un-burned-in simulator. The repair became the floor standard used in every later experiment: the z-margin gate of Section 2 (a score must clear its null mean by three null standard deviations, so hair-thin residuals over an unstable floor cannot pass), a burn-in rule (the first 2,000 steps of every trace were discarded, so a simulator's settling-in transients could not masquerade as structure), and a reproducibility fix for the older simulator (its traces were cross-process reproducible only with a pinned hash seed, without which floors and scores can silently come from different worlds). A residual 30% theft by a genuinely moving meter is real Granger flow with no causation (it helps predict without causing anything): diffuse shared-history witnessing that no single conditioning key can screen. This became a benchmark requirement for the conventions of Experiment 2.

5 Experiment 2: which conventions survive more complex worlds?

Experiment 2 asked which measurement conventions survive outside the world they were built for. Two new conventions were first qualified against the full anchor criteria before entering the benchmark. *Fused + grown keys* exposed its own limit during its justifying unit test: greedy growth cannot climb a flat plateau (in pure three-way parity every single key has zero marginal gain) while excelling on graded composition (majority rules, +43% over best-key); it refused GP 100% of the time, falsifying our prediction it would inherit best-key's witness blindness, because *procedure-mirroring floors self-penalize search depth*: a deeper maximization must find real structure to outrun its own inflated null. The two block-level conventions also qualified (100% on every criterion; the block context absorbed even the chance blips seen elsewhere).

Fig. 1 summarizes the results of this experiment: goal-identification rates across the four worlds under every convention. **Any-block identified every planted goal in every world**: the only convention that never missed. Mechanistically, 'target block's past plus one key element' is two-block conditioning, which decrypts three-way composition and, crucially, does not require the goal to belong to a detected agent: since the detector orphans goals, this robustness is fitness for the terrain. Own-block, conditioning only on the candidate's own block, has no ring for an orphaned goal (0% on the parity goal; 70% worst-goal in the coupled colony); the divergence between the two thus became our partition-quality diagnostic. Every variable-level convention hit a scale wall. Fused's wall is combinatorial: the state count doubles per variable. Fused + best-key and fused + grown keys hit theirs on the intake side, which saturates outright at 49 variables (19,999 distinct joint states in 20,000 samples), so intake reads as bias for every variable. The *no-core control caught* this defect (0%

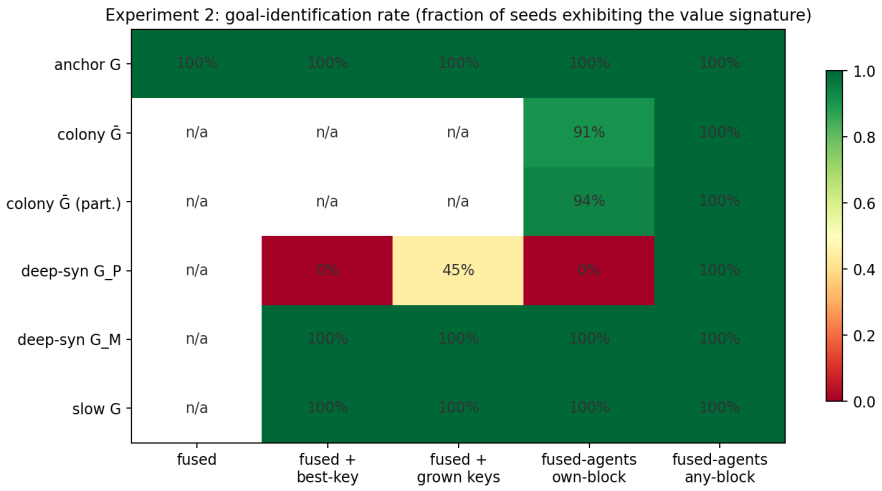


Figure 1: Experiment 2: fraction of seeds in which each planted goal exhibits the value signature, per convention (columns) and world (rows). 'n/a' marks recorded infeasibility (fused beyond ~ 10 variables; fused + best-key and fused + grown keys intake at colony scale). Any-block is the only column without a miss.

clean seeds), and the locked interpretation rules converted it into a recorded infeasibility rather than a spurious result. The synergy cliff was measured on the parity goal (best-key 0%, fused + grown keys 45%, any-block 100%; the majority goal 100% for all), and re-running colony failures with coupling removed (our pre-registered contingency) attributed both own-block's shortfall (70% $\rightarrow$ 90%) and the single recurring false positive (an action wire, 6/20 $\rightarrow$ 2/20) to ring interference, not scale.

6 Experiment 3: how easily can the instrument be fooled?

Experiments 1–2 planted values and asked whether they could be found; Experiment 3 planted impostors and asked whether they could fool the surviving conventions (any-block primary, own-block diagnostic), in colony-style worlds, with or without an interventional yardstick.

The yardstick. The idea is to restate the value signature under interventional access: a genuine value 'levers the world, unlevered by the world'. The UAD benchmark supplied the ingredients. It embeds a `goal_flip` operation, a 'handle': an operation available from within the world, rather than an idealized intervention imposed from outside it. Applied to the true goal it changes the action within the tick; applied to anything else it does nothing. We generalized it to a *do-style flip* of any variable's value as consumed by the dynamics, on 50% of interleaved steps, and built the two-direction *interventional mirror* of the value signature: a *goal-role probe* (does flipping X change its agent's behaviour, given the agent's policy context?) and an *autonomy probe* (does flipping X's own environment change X?). The interventional verdict required a role effect above floor *and* an autonomy effect at floor. Its construction was itself instructive: the XOR cipher bit three times (an unconditioned role probe; a saturating autonomy readout; unconditioned autonomy terms) and was caught each time by a validation bench on the anchor, where the do-style flip had to agree with the embedded operation before use; the fix, each time, was the context conditioning the benchmark's own interventional score already employed.

Table 4: Experiment 3: the limits of passive detection, per impostor, and what interventional access adds; 20 seeds; passive = any-block.

Impostor	Best passive verdict	Yardstick verdict	Access adds
Fast captured goal	refused (20/20)	refused (20/20)	none
Slow captured goal	**wrongly identified (20/20)**	refused (20/20)	**total**
Twin, 0–0.5% noise	twin wrongly identified (19–20/20)	twin refused (20/20)	**total**
Twin, $\geq 2\%$ noise	twin refused (19–20/20)	twin refused (20/20)	none

Results. Table 4 summarizes the results of this experiment: for each impostor, the best passive verdict, the interventional verdict, and what interventional access adds over the best passive method, which together map the limits of passive detection. The fast captured goal (wired into the action but copying its own patch) was refused in 20/20 seeds: its intake is large and visible. Its audit also recorded an emergent effect: copying the patch its agent's belief copies, it partially cancels the belief and *pacifies its own environment*. The slow captured goal (flipping at a true goal's tempo, 1.77%, with every flip verified to follow a six-step run of its patch) **was wrongly identified by the value signature in 20/20 seeds**. Its environment-intake was above its noise floor at lag 1 (net ≈ 0.03 nats); however, its drive-to-intake ratio exceeded the value signature's leak-tolerant clause (Section 2). The impostor's drive-to-intake ratios spanned 10.3–15.0 (our threshold was 9), so thresholds modestly above this range refuse it in every seed, while the lowest ratio any true goal exhibited was 125. The vulnerability thus belongs to the leak-tolerance choice, not to passive observation itself: a stricter gated passive test refuses this impostor at lag 1 (see Section 8).[1] Within the value-signature instrument itself, multi-horizon conditioning remains the evidenced remedy (the intake also surfaces above the lag-1 floor at lag 2 in 20/20 seeds). The yardstick unmasked it in 20/20 (role yes, autonomy yes). Fig. 2 gives the passive breaking point for the goal role. Twins are indistinguishable from their goals at 0–0.5% noise, as they must be: identical sequences cannot be separated by any passive method. They are transitional at 1% and separated at $\geq 2\%$ (19–20/20), with no inversions. The interventional yardstick, by contrast, refused the bit-identical twin even at 0% (a recording levers nothing) while its goal kept the verdict. Interventional access resolves both cases; only the zero-noise twin is known to *require* it.

The twin channel. In the alias no-core condition the *disconnected* goals still exhibited the value signature in 20/20 seeds, and correctly so: a goal stripped of its behavioural wiring still causally drives its own recording, so 'drives without being driven' genuinely holds through the record. The instrument was right and the control's expectation was mis-specified for twin-bearing worlds. Philosophically the point is sharp: a value can persist in the value signature through its records alone.

7 Discussion

What was established. The value signature locates a planted value structure from passive observation across single- and multi-agent worlds, and refuses decoys, disconnected goals, scrambles, and (with the right conditioning) meters; the block-level any-block convention is the only tested instrument that holds at the scale of Zarncke's multi-agent simulator. Its limits are now measured

[1] Our pre-registered reading was different: we expected this misidentification on the ground that one step of conditioning cannot see a trigger living in run history, and at the time took the result to show exactly that. The comparison of Experiment 5 falsified that reading.

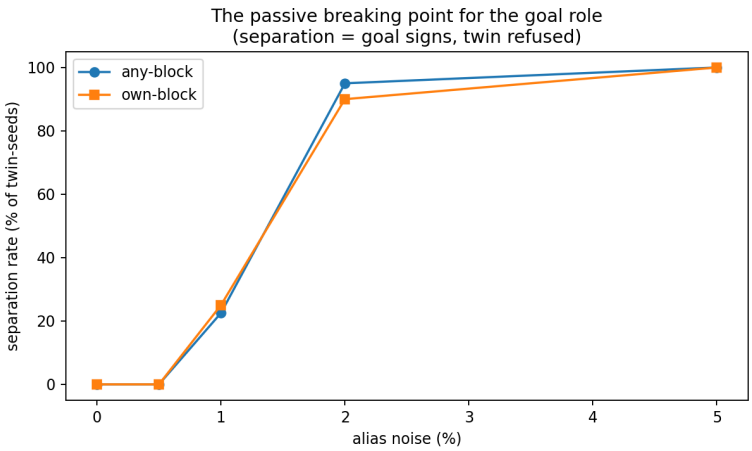


Figure 2: Experiment 3: separation rate (the goal exhibits the value signature, the twin does not) versus twin noise: the passive breaking point for the goal role lies between 1% and 2%.

rather than conjectured: a history-triggered captured goal can only be passively refused if a sufficiently high value is given to the value signature's leak-tolerant intake clause; however, twins below $\approx$ 1–2% noise are passively indistinguishable from goals. Using an interventional yardstick closes this gap, giving a quantitative form of the access-model thesis of Zarncke [8]. Several findings concern method rather than result: the XOR cipher; the collapse-detector role of the no-core control; and the self-penalizing floors.

Why structure, not behaviour. Behavioural inference and structural discovery are complementary, but structure may have a distinctive advantage under adversarial conditions: an agent that conceals its objectives by adding decision noise weakens the behavioural signal, yet the component that most drives behaviour beneath that noise remains structurally locatable; once located, it can be isolated and studied in a regime where the noise no longer obscures it. Our results are a first indication that such structural location is possible.

Limitations. (1) *Single-variable values.* The instrument scores individual variables. It has been tested on agents with multiple belief states but not with multiple goal or value states; a value structure distributed over several components is beyond it as it stands. The fused + grown keys results suggest one route (deciphering interactions among several value-like variables); we conjecture a better one: use UAD's boundary-detection machinery to identify sub-systems within an agent, then evaluate the empowerment and plasticity of those sub-systems rather than of single variables. (2) *Lag-1 conditioning together with the leak-tolerant intake clause*, which the slow captured goal jointly exploits; multi-horizon conditioning and a stricter stability gate are both evidenced remedies (the latter from the comparison of Experiment 5); what strictness costs on genuinely leaky value cores is untested. (3) *Planted structure.* All worlds are engineered; whether the polarization *emerges* in evolved or learned systems is the question the instrument now makes askable, not one it has answered. (4) The protocol was soft-blind: the implementing system had read the world's source; protection came from the world's independent origin, pre-registration, controls, and multi-seed sweeps. (5) The relation between this signature and Omohundro's drives is a hypothesis supported by recoverability, not a demonstration that empowered-and-stable components *function* as values in richer agents.

8 Further work

Two further experiments were conducted after the work reported above and inform some corrections woven into this text, as well as suggesting possible lines for future inquiry.

8.1 Experiment 4: an event-log deployment world

Experiment 4 used the instrument in a different kind of world: an event-log simulation of a deployment pipeline, also developed by Zarncke,[2] containing agents performing different tasks and carrying an injected self-preserving utility. The experiment showed that it was possible to deploy the instrument in such an environment. The instrument identified the value structure footprint not in the outcome state the utility rewards (which was strongly driven by the agents' environment) but in the drive-without-intake of the actions the utility selects for (regime-graded, and visible in the auditor's log alone); in event-log worlds the mark of a value is on the choices, not the scoreboard.[3]

8.2 Experiment 5: an independently derived formalism compared

Our final experiment ran our instrument side by side with an interventionist, role-sufficiency-based alternative independently derived by Jobst Heitzig (roles as minimal internal representations sufficient for named external targets).[4] The comparison converged wherever the formalisms overlap (including an interventional battery identical to our yardstick) and diverged instructively twice, in mirror image: a strict stability gate (theirs) passively unmasked the slow captured goal our original leak-tolerant clause admitted, while an evidence-only stability quantifier (theirs) wrongly identified the witness class of causally inert meters as goal-like, where our all-source intake refuses them. A new world with the value structure *outside* the agent it drives was identified by the value signature in 20/20 seeds while remaining invisible to any internal-only candidate scope; and on observation-only data the sufficiency tests cannot be stated at all, while the passive value signature runs. On this evidence, the load-bearing difference between the two routes to the same construct is the active-vs-passive contrast itself. It decides what needs a do-operator, what passivity recovers, and what can even be asked of data that only watches.

8.3 Possible lines for future inquiry

We identified the following as potentially promising lines of further experimentation.

- *Systems-of-agents environments.* Worlds in which a value structure is held by a coalition rather than a single agent. The any-block convention makes no assumption about which agent owns a value (any block can serve as the key that reveals any candidate's drive), so the instrument is structurally ready; what is missing are environments that actually plant coalition-held values to identify.
- *Multi-horizon conditioning.* The slow captured goal's trigger surfaces above the lag-1 floor at lag 2 in 20/20 seeds, so extending the conditioning window beyond one step is an evidenced upgrade rather than a speculation; the design question it opens is how to keep the noise floors procedure-mirrored and honest as the window (and with it the state space) grows.

[2] https://towards-alignment.com/cards/experiment/deployment-pipeline-simulator/
[3] https://github.com/SJ-Beard/deployment-pipeline-value-detect
[4] https://github.com/SJ-Beard/active-v-passive-value-detect

- *Sub-system-level scoring.* Score discovered sub-systems rather than single variables, using the UAD boundary machinery *inside* an agent to propose components and then evaluating each component's empowerment and plasticity, to reach value structures distributed over several variables, which the present instrument cannot represent.
- *The twin channel as a persistence probe.* A goal stripped of its behavioural wiring still drives its own recording, so the value signature survives through records (found in Experiment 3's no-core control). Deliberately exploiting this would probe how long, and through what kinds of record, a value structure remains locatable after it stops steering behaviour.
- *Testing a stricter instrument on genuinely leaky values.* Experiment 5 showed that a strict stability gate catches the captured goals the leak-tolerant clause admits, while our all-source intake refuses the inert meters an evidence-only gate admits; combining the two could give the best instrument yet. The experiment worth running would plant *genuinely leaky* true values (values with real, measurable environmental intake, which the leak-tolerant clause was designed to protect) and measure what the stricter setting costs: whether some threshold refuses every captured goal while keeping every leaky-but-real value, or whether the two demands are in unavoidable tension.

Reproducibility

All code, world generators, audits, locked pre-registrations, verdict tables, and figures are available, with fixed seeds, from `https://github.com/SJ-Beard/value-detect`. The UAD benchmark code is imported unmodified from `https://github.com/GunnarZarncke/agency-detect`. Sweeps ran on a single computer (Experiment 1 $\approx$ 1 h; Experiment 2 $\approx$ 12.5 h; Experiment 3 $\approx$ 37.5 h at four workers).

Statement on use of AI

The author conceived the hypothesis and framing, made all design decisions, reviewed every audit and pre-registration in plain language, and directs the programme. Claude (Anthropic), an AI system, wrote all code, ran the experiments and analyses, and drafted this manuscript under the author's direction; every decision, deviation, and defect is dated in the project's decisions log. **This manuscript is, as it stands, AI-generated**: it was written by Claude and revised and approved by the author, who intends to write the account in their own words when time allows. The author takes full responsibility for the content. The manuscript is not being submitted for publication.

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